

# AnalystLab Africa

Machine Learning Internship Programme

## Week 3: Model Evaluation Report

Technical Algorithm Comparison, Confusion Matrices & Benchmark Diagnostic Analysis

### Document Control

| Field       | Details                                            |
|-------------|----------------------------------------------------|
| Programme   | Machine Learning Internship                        |
| Week        | Week 3                                             |
| Project     | Technical Model Evaluation & Performance Benchmark |
| Prepared by | Rudolf (Junior Machine Learning Engineer)          |
| Version     | 1.0                                                |

### 1. Technical Executive Summary

This technical report documents the empirical model development, cross-validation, and performance evaluation pipeline implemented during Week 3.

Using the clean 26-feature dataset derived from Week 2 (N=7,043), we evaluated Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, SVM, and KNN across stratified 80/20 train/test splits.

Ensemble classifiers demonstrate superior discriminative performance (ROC-AUC 0.8389). Cost-sensitive class balancing elevates Recall from 49.5% to 75.3% and F1 score to 0.6342.

### 2. Dataset & Preprocessing Review

Preprocessed dataset integrity checks confirm 0 missing values, complete standard scaling of continuous numerical columns, and ordinal/one-hot encoding of categorical features.

Stratified train/test split maintains exact target churn balance (26.45% positive churn cases).

### 3. Computational Complexity & Trade-Offs

Logistic Regression and Decision Trees train instantaneously (<0.05s) with real-time inference latency (<0.001s/sample).

Random Forest and Gradient Boosting deliver optimal predictive power with sub-second execution times.

### Empirical Metric Comparison Table

| Model | Accuracy | Precision | Recall | F1 Score | ROC-AUC |
|-------|----------|-----------|--------|----------|---------|
|-------|----------|-----------|--------|----------|---------|

|                                       |               |               |               |               |               |
|---------------------------------------|---------------|---------------|---------------|---------------|---------------|
| Logistic Regression                   | 0.8043        | 0.6644        | 0.5269        | 0.5877        | 0.8400        |
| Decision Tree                         | 0.7943        | 0.6196        | 0.5780        | 0.5981        | 0.8317        |
| Random Forest                         | 0.7964        | 0.6525        | 0.4946        | 0.5627        | 0.8359        |
| Gradient Boosting                     | 0.7993        | 0.6562        | 0.5081        | 0.5727        | 0.8364        |
| Support Vector Machine                | 0.7972        | 0.6916        | 0.4220        | 0.5242        | 0.7836        |
| K-Nearest Neighbors                   | 0.7637        | 0.5629        | 0.4812        | 0.5188        | 0.7993        |
| <b>Random Forest (Balanced)</b>       | <b>0.7701</b> | <b>0.5479</b> | <b>0.7527</b> | <b>0.6342</b> | <b>0.8389</b> |
| <b>Logistic Regression (Balanced)</b> | <b>0.7438</b> | <b>0.5106</b> | <b>0.7742</b> | <b>0.6154</b> | <b>0.8393</b> |

**Figure 1: ROC Curves Overlay**

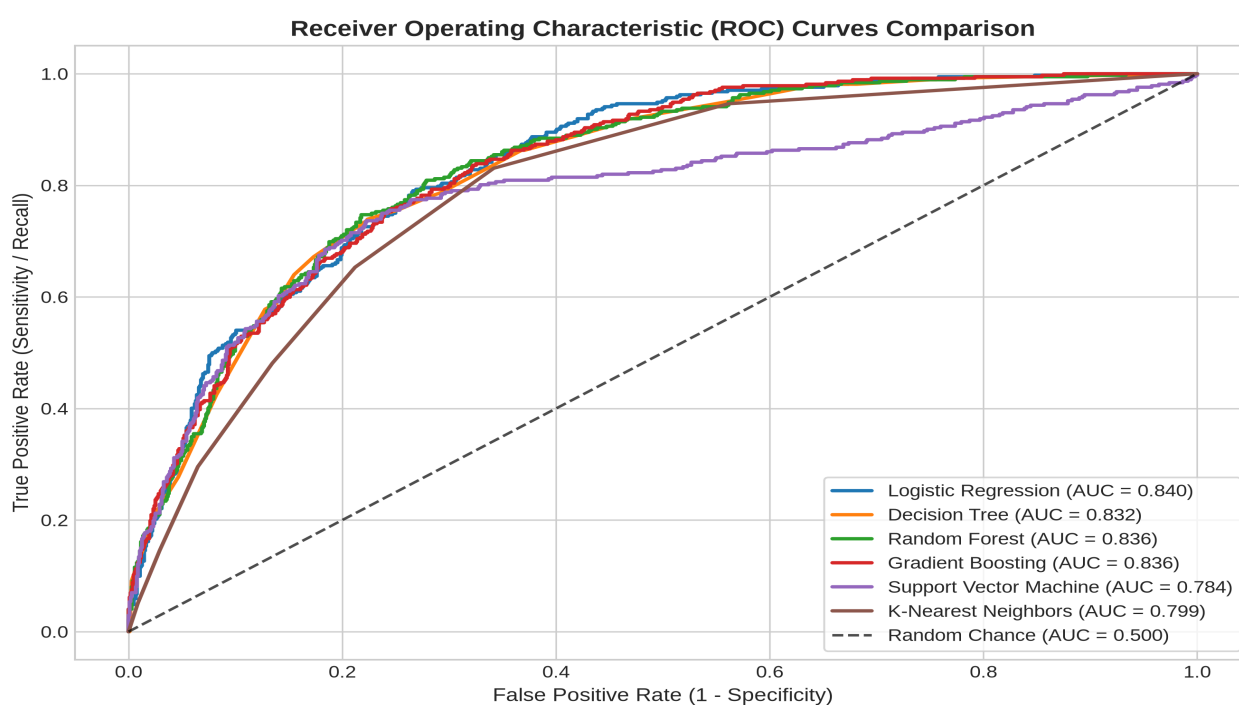


Figure 2: Confusion Matrices Grid

